Boundary & Edge Case Testing

This is Sample Document 3, used in boundary tests that upload three PDF files simultaneously to the Merge PDF tool. Three-file merges represent a practical boundary for most users and help validate that the tool handles multiple uploads without degradation.

Boundary value analysis (BVA) identifies input values at or near the edges of equivalence partitions. For file upload features, relevant boundaries include: minimum files (1), typical files (2-3), and maximum files (20+). This framework currently covers the 1, 2, and 3-file boundaries.

All three sample PDF files are structurally identical in format but carry distinct metadata, titles, and content so that the merged output is clearly verifiable as a combination of three separate documents.

Data Table

Boundary	Value	Test ID	Expected
Min upload	1 file	TC_MERGE_004	Upload succeeds
Typical	2 files	TC_MERGE_005	Both listed
Upper boundary	3 files	TC_MERGE_006	All listed
Merge 2	2 files	TC_MERGE_008	Success / download

Test Environment Configuration

The framework reads all environment configuration from src/test/resources/config/config.properties. Key properties include base.url, browser, headless, implicit.wait, explicit.wait, page.load.timeout, and retry.count.

WebDriverManager (Bonigarcia) handles ChromeDriver, GeckoDriver, and EdgeDriver binary management automatically, eliminating manual driver version synchronisation. The framework supports Chrome, Firefox, and Edge via the BrowserType enum and DriverFactory switch statement.

For CI/CD pipelines, set headless=true and pass the browser parameter via the TestNG XML suite or Maven command line: -Dbrowser=chrome -DheadlessMode=true

Data Table

Property	Default	CI Override	Purpose
browser	chrome	chrome	Browser type
headless	false	true	Headless mode
base.url	ilovepdf.com	ilovepdf.com	Target URL
retry.count	2	2	Flaky test retries
thread.count	3	5	Parallel threads

Continuous Integration Pipeline

This framework is designed to integrate with any CI/CD system that supports Maven. The standard pipeline stage is: mvn test -DsuiteXmlFile=src/test/resources/suites/smoke.xml for the smoke gate, followed by the regression suite on success.

Test reports are archived as build artifacts. The ExtentReports HTML file and all failure screenshots are stored under reports/ and screenshots/ respectively. These directories should be added to the CI artifact collection step.

Parallel execution is configured at the suite level in the TestNG XML files. Set parallel='tests' or parallel='methods' and adjust thread-count based on available CI runner resources. ThreadLocal in DriverManager ensures each thread has its own isolated WebDriver instance.

Data Table

Stage	Suite	Gate	On Failure
PR Check	smoke.xml	Block merge	Notify author
Post-merge	regression.xml	Block deploy	Alert team
Nightly	e2e.xml	Advisory	Log ticket
Release	all suites	Block release	Escalate